

**Aggregate Insider Trading and the Predictability of Market Returns:
Contrarian Strategy or Managerial Timing?**

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Abstract

We decompose realized market returns into expected return, unexpected cash flow news and unexpected discount rate news to test the relation between aggregate market returns and aggregate insider trading. Our motivation is to distinguish whether the observed relation between market returns and insider trading is due to contrarian strategy or managerial timing. We find that (1) the predictive ability of aggregate insider trading is much stronger than what was reported in earlier studies (2) aggregate insider trading is strongly related to unexpected cash-flow news and (3) market expectations do not cause insider trading, contrary to what others have documented. These results strongly suggest that the predictive ability of aggregate insider trading is because of managerial timing rather than contrarian strategy. These results hold even after we control for information uncertainty by using firm size as proxies.

Aggregate Insider Trading and the Predictability of Market Returns: Contrarian Strategy or Managerial Timing?

1. Introduction

Recent studies on aggregate insider trading have documented that insiders are able to predict future market movements and that they are able to time the market (Seyhun (1988), Lakonishok and Lee (2001)). However, it is not clear from the evidence whether this predictability of market returns is due to insiders being contrarian investors (Rozeff and Zaman (1998), Lakonishok and Lee (2001)) or whether managers are better informed about their firm's future prospects and it is this information that explains their market timing ability (Ke, Huddart and Petroni (2003)) or whether it is a function of both (Piotroski and Roulstone (2005)).

There is substantial evidence that corporate officers and directors are able to discern apparent mispricing in their firm's securities based on firm related information and are able to profitably trade on this.¹ If this information is related to future economy wide activity then aggregate insider trading should predict future market movements and the market timing ability of insiders would be based on information unanticipated by the market (see Seyhun,1988) . We differentiate this from the contrarian investment strategy of insiders and define it as managerial timing. If insiders are motivated to trade because of perceived mispricing, it is also conceivable they may react to market returns. It is possible that noise traders may

¹ Previous studies based on US data unanimously documented that insiders are better informed and earn abnormal returns [Lorie and Niederhofer(1968), Jaffe(1974), Seyhun(1986), Rozeff and Zaman(1988), Lin and Howe(1990) and Lakonishok and Lee(2001)]. Using Oslo Stock Exchange data Eckbo and Smith(1998) show that insiders do not earn abnormal returns while Jeng, Metrick and Zeckhauser(2003) show that abnormal returns earned by insiders are restricted only to purchases.

drive market prices way from intrinsic values even in the absence of new information. Hence, a stock that was trading roughly at its intrinsic value could decline (rise) significantly because of such noise trading. Corporate insiders may then perceive the stock to be undervalued (overvalued) and buy (sell) it. To the extent that noise trading is a marketwide phenomenon, we would then expect market returns to 'predict' aggregate insider transactions (See Rozeff and Zaman (1998), Chowdhury, Howe and Lin (1993) and Lakonishok and Lee (2001). Such a relationship would be viewed as insiders following a contrarian investment strategy. Even though under both strategies insider trading is related to market returns, the key distinction is that managerial timing implies insider trading will predict future market returns while contrarian strategy implies insider trading is a reaction to market returns

Other related studies of managerial decisions also suggest that insiders are better informed about their companies' future prospects. For example, Ikenberry, Lakonishok and Vermaelen (1995) find positive abnormal returns earned by shareholders of companies that have announced open market share repurchases. These abnormal returns persist for some time after the announcement. One of the main motivations for repurchases seems to be that insiders perceive the company's stock as being undervalued. Loughran and Ritter (1995), on the other hand, observe a prolonged underperformance by companies following seasoned equity offerings. This is in line with the hypothesis that companies tend to issue seasoned equity when they perceive the market to be too optimistic about the prospects of their company. Baker and Wurgler (2000) find that the share of equity issues in total new equity and

debt issues has been a stable predictor of U.S. stock market returns between 1928 and 1997 (see Baker, Taliaferro and Wurgler, 2006 among others).

A related line of research on insider trading has focused on whether aggregate insider trading can predict market movements and could be used as a tool to time the market. Seyhun (1988, 1992) provides evidence suggesting that some of the mispricing observed by insiders in their own firms' securities is caused by unanticipated changes in economy wide activity. He finds that aggregate insider transactions are correlated with the return on the market during the subsequent two months.

Chowdhury, Howe and Lin (1993) find that stock market returns Granger-causes insider transactions while the predictive content of aggregate insider transactions for subsequent market returns is slight. Lakonishok and Lee (2001) also provide evidence in support of the predictive ability of aggregate insider trading and market movement. They conclude that this ability is partially explained by their finding that insiders act as contrarian investors.

Previous studies simply examine the relationship between realized market return and some metric of insider trading without explicitly considering the source of predictability. Piotroski and Roulstone (2005) is an exception. They are the first try to differentiate the source of the predictability. However, they use the change of account returns as proxies for future cash flows. Cohen, Gompers, and Vuolteenaho (2002) point out that the change in account returns is not suffice to proxy future cash flows. Both conclusions of contrarian strategy of investing by insiders and managerial timing rely on insider trading to be positively related to subsequent

realized market returns. However, these studies make no attempt to determine whether the apparent predictability of market returns by aggregate insider trading is due to contrarian strategy or managerial timing. The purpose of this paper is to re-examine the ability of aggregate insider trading to predict market-wide movement using return decomposition in a vector autoregressive (VAR) model framework. Such a re-examination is **a** called for because of mixed results reported in previous papers. In order to distinguish between these two sources of predictability we closely follow Campbell (1991) and Hecht and Vuolteenaho (2005) method of decomposing aggregate market return into expected return, unexpected cash-flow news and unexpected discount rate news. We argue that managerial timing suggests a positive relation between aggregate insider trading and unexpected cash-flow news while contrarian strategy would suggest a negative relation between insider trading and expected return. Using this decomposition, a regression of market returns on insider trading measures is then decomposed into three component regressions. We find the following: (1) the predictive ability of aggregate insider trading is much stronger than what was reported in earlier studies (2) aggregate insider trading is strongly related to unexpected cash-flow news and (3) market expectations do not cause insider trading, contrary to what others have documented. These results strongly suggest that the predictive ability of aggregate insider trading is because of managerial timing rather than contrarian strategy.

Our contribution is two-fold. First, this paper provides definitive evidence into the debate of whether insider trading based on perceived mispricing is a result of contrarian investment strategy or whether it is based on insiders' access to

information about future cash flow news. By decomposing realized market returns into expected returns, unexpected cash flow news and unexpected changes in discount rate this paper directly tests the sources of the insider trading predictability. Second, this paper contributes to the existing literature on the importance of the relation between corporate transactions and insiders ability to time the market. When the firm's securities are mispriced and insiders are able to identify this mispricing, then this ability affect the financing, investment and other corporate transactions.

The paper is organized as follows. Section 2 describes the methodology used while section 3 describes the data and provides summary statistics. Results are reported and discussed in section 4 while the last section contains a summary and interpretation of the results.

2.The information content of aggregate insider trading: managerial timing, or contrarian strategy?

There are a number of compelling and competing reasons to believe that aggregate insider trading can predict future market returns. Assume that company executives and directors know their businesses more intimately than analysts (investors) following their stocks. They know when demand for their goods and services are increasing, when inventories are piling up, when production costs are increasing or profit margins declining, etc. Given their knowledge about their firm, insiders should be able to predict, say, if the firm's future cash flows would increase and would buy stocks in their firms. If the cash flows are related to economy wide activity such as increases in aggregate demand of goods and services then subsequently when the

increase in economy wide activity is recognized by the market, stock prices will rise. This will result in a positive relation between insider trading and market return. On the other hand, if the predicted increase in cash flows by insiders is strictly the result of some firm-specific improvement (e.g. profit margin) there should be no relation between insider trading and market return. We call this the managerial timing hypothesis. Seyhun (1988, 1998) is the first study that documents aggregate insider trading is positively related to market activity and in a later paper provides evidence of aggregate insider trading being related to macro-economic variables.

A competing hypothesis regarding aggregate insider trading relies on the contrarian strategy of investing. If stock prices are affected by the trading of both informed and uninformed (noise) traders then prices can diverge from fundamental values (Shiller 1984, De Long et al 1990). According to this view noise traders may drive market prices away from current fundamental values. However, in the long run prices would revert back to fundamental values. If the contrarian strategy is employed by insiders at the firm specific level then there should be no relation between market returns and insider trading. On the other hand, if 'noise' trading is a market wide phenomenon then a relation between aggregate insider trading and market return should exist. In such a scenario, market returns would 'predict' insider trading behavior. Chowdhury, Howe and Lin (1993) and Lakonishok and Lee(2001) provides evidence in support of aggregate insider trading being driven by the contrarian strategy.

3. Framework and hypotheses

In order to test the relation between aggregate stock returns and inside trading and whether the relation is due to the contrarian strategy or managerial timing we use the standard log-linear approximation of present value model developed by Campbell (1991).

A. Log-linear present value model framework and insider trading

Campbell (1991) decomposes the realized return on equities into following three components:

$$\begin{aligned} R_{t+1} &= E_t R_{t+1} + (E_{t+1} - E_t) \sum_{j=1}^{\infty} \rho^j \Delta D_{t+1+j} - (E_{t+1} - E_t) \sum_{j=1}^{\infty} \rho^j R_{t+1+j} \\ &= E_t R_{t+1} + N_{CF,t+1} - N_{DR,t+1} \end{aligned} \quad (1)$$

where R is the log return on equities, ΔD is dividend growth, ρ is the discount factor, $E_t(R_{t+1})$ is the one-period expected return, $N_{CF,t+1}$ is the cash flow news, and $N_{DR,t+1}$ is the discount rate news. This equation states that the realized return must be associated with the expected return, the changes in expectations of future cash flows, and/or the changes in the expectations of future discount rates. As emphasized by Campbell (1991), equation (1) is really nothing more than a dynamic accounting identity relating the current return innovation to revisions in expectation.

Hecht and Vuolteenaho (2005) apply this method to measure the relative importance of these three effects in regressions of returns on cash flow proxies. Based on the equation (1), the explanatory power of cash flow proxies may arise from the correlation of cash flow proxies (predictors) with one-period expected returns, cash flow news, and/or expected return news. They argue that “If expected-return variation is responsible for the high explanatory power of the aggregate regressions, these R^2 should not be interpreted as evidence of cash-flow news driving the returns.

Similarly, if expected-return news is highly variable and positively correlated with cash-flow news, the low R^2 s in regressions of firm-level returns on earnings do not necessarily imply that earnings are a noisy or delayed measure of the cash-flow-generating ability of the firm. Even if earnings are a clean signal of cash-flow news, expected-return effects (due to variation in risk-adjusted discount rates and/or mispricing) can garble the earnings-returns relation.”

In a similar spirit, we apply Campbell’s decomposition to estimate and test the dynamic relation between aggregate market returns and aggregate insider trading. This method helps us to distinguish the whether relation between aggregate market return and insider trading is due to a contrarian strategy or managerial timing. Consider a typical forecast regression of returns on insider trading:

$$R_{t+1} = \alpha + \beta IT_t + e_{t+1} \quad (2)$$

where IT is a measure of insider trading. Seyhun (1988) uses similar methodology to show a weak relationship between insider trading and market returns and concludes insider transaction predict market return. As analyzed above, it is difficult to interpret the coefficient β , and more importantly, using regression (2) we can not distinguish whether the relation between aggregate market returns and insider trading is due to the contrarian strategy or managerial timing.

Using Campbell’s (1991) decomposition, however, we can rewrite the regression (2) as following:

$$\begin{aligned} E_t R_{t+1} &= \alpha + \beta_{ER} IT_t + e_{ER,t+1} \\ N_{CG,t+1} &= \alpha + \beta_{CF} IT_t + e_{CF,t+1} \\ -N_{DR,t+1} &= \alpha + \beta_{-DR} IT_t + e_{-DR,t+1} \end{aligned} \quad (3)$$

Since the sum of the left-hand-side in regression (3) is the realized return and the independent variable in regression (3) are same, regression (2) can also be expressed as:

$$R_{t+1} = \alpha + (\beta_{ER} + \beta_{CF} + \beta_{DR}) IT_t + (e_{ER,t+1} + e_{CF,t+1} + e_{DR,t+1}) \quad (4)$$

Regression (3) and (4) show that there are three sources driving the relation between aggregate market return and insider trading: one-period expected return, cash flow news, and discount rate news.

We also consider the following regression:

$$\begin{aligned} IT_{t+1} &= \alpha + \gamma R_t + u_{t+1} \\ IT_{t+1} &= \alpha + \gamma_{ER} E_t(R_{t+1}) + u_{ER,t+1} \\ IT_{t+1} &= \alpha + \gamma_{CF} N_{CG,t} + u_{CF,t+1} \\ IT_{t+1} &= \alpha + \gamma_{DR} (-N_{DR,t+1}) + u_{DR,t+1} \end{aligned} \quad (5)$$

Equation (4) shows, if expected-return variation is responsible for the high explanatory power of the aggregate regressions, these R^2 should not be interpreted as evidence of managerial timing driving the returns. Similarly if expected-return news is highly variable and positively correlated with cash-flow news, the low R^2 s in regressions of market returns on inside trading do not necessarily imply that insider trading is a noisy or delayed measure of the cash-flow-generating ability of the firm. Even if insider trading is a clean signal of cash-flow news, expected-return effects (due to variation in risk-adjusted discount rates and/or mispricing) can garble the insider trading-returns relation. We use regression (3) and (5) to estimate the relation between aggregate market return and insider trading, and distinguish whether the relation is attributed to the managerial timing (as evidenced in Seyhun (1988)) or contrarian strategy as claimed by Chowdhury, Howe and Lin (1993) and Lakonishok and Lee(2001).

Managerial timing implies that insiders are better able to predict future cash flow news of the firm than outside investors. If these cash flows are related to economywide activity then subsequent to aggregate insider buying (selling) in stocks of their firm the aggregate market returns should increase (decrease). If the hypothesis of managerial timing is true then we expect a positive and significant coefficients for β_{CF} and β_{DR} . In contrast, if managers do not have the ability to time the market the coefficients β_{CF} and β_{DR} will be insignificant. Furthermore, if our conjecture of managerial timing is that insider managers know more about their cash flow news and aggregate insider cash flow news do not cancel out but rather are proxies of aggregate market cash flow news then the coefficient β_{CF} should dominate β_{DR} .

Contrarian strategy implies that outsider investors make valuation errors through the application of inferior valuation models and/or the incorporation of biased judgments. Based on the perceived mispricing, insiders trade against outside investors' sentiment. If the contrarian strategy drives the relation between aggregate market return and insider trading, we will expect that γ_{ER} to be significantly negative. For instance, if outside market expectation $E_t[R_{t+1}]$ is positive, and if inside traders perceive that this expectation is wrong, insider traders will sell their stocks, i.e., γ_{ER} is negative. In this case note that the co-efficients of cash flow news and discount rate news should be insignificant.

Hypothesis 1: If insider trading is not informative (in terms of managerial timing) then the lagged values of β_{CF} , β_{DR} are indistinguishable from zero; otherwise they are positive and significant.

Hypothesis 2: If insider trading is not informative (in terms of contrarian strategy) then the lagged values of γ_{ER} are indistinguishable from zero; otherwise they are negative and significant.

B. Estimating one-period expected returns, cash flow news and discount rate news

We follow Campbell (1991), and Campbell and Vuolteenaho (2004) to estimate the one-period expected return, cash flow news, and discount rate news series using a vector autoregressive (VAR) model.

We assume that the data are generated by a first-order VAR model

$$Z_{t+1} = A_0 + AZ_t + u_{t+1} \quad (6)$$

where Z_{t+1} is a vector of excess log market returns, the term yield defined as the yield difference between ten-year constant-maturity taxable bonds and short-term taxable notes, the price-earnings ratio from S&P 500 index, and small-value spread², describing the economy at time $t+1$, A_0 and A are vector and matrix of constant parameters, and u_{t+1} is a vector of shocks. With the VAR expressed in this form, the components of identity (1) can be obtained by

$$\begin{aligned} E_t R_{t+1} &= eI'(A_0 + AZ_t) \\ N_{CG,t+1} &= [eI' + eI'\rho A(I-\rho A)^{-1}]u_{t+1} \\ -N_{DR,t+1} &= eI'\rho A(I-\rho A)^{-1}u_{t+1} \end{aligned} \quad (7)$$

where $eI' = [1 \ 0 \ \dots \ 0]$, and I is an identity matrix. Equation (7) expresses $E_t R_{t+1}$, the one-period expected return as fitted value of Z_{t+1} based on VAR model in equation (3),

² For details of data construction, see Campbell and Vuolteenaho (2004)

$N_{CG,t+1}$, the cash-flow news, and $N_{DR,t+1}$, the discount rate news as linear functions of the $t+1$ shock vectors..

4. Data and Summary Statistics.

A. VAR data

In order to decompose the realized return into expected return, cash flow news and discount rate news using VAR approach, we need to specify variables to be included in the state vector. Following Campbell and Vuolteenaho (2004), we choose a model with the following four state variables: the excess market return (measured as the log excess return on the Center for Research Security Prices [CRSP] value-weighted index over Treasury bills; the term yield spread between long-term and short-term bonds (measured as the difference between ten-year constant-maturity taxable bond yield and the yield on short-term taxable notes); the market's price-earnings ratio (measured as the log ratio of the S&P 500 price index to a ten-year moving average of S&P 500 earnings); and small-stock value spread (measured as the difference between the log book-to-market ratios of small value and small growth stocks). Asset pricing literature finds that these state variables are able to forecast and track aggregate market returns³.

B. Inside trading data

We collect all insider trading information from the Securities Exchange Commission (SEC) Ownership Reporting System (ORS). The ORS data starts in 1975 and ends in 2000 and contains all insider transaction data that are subject to disclosure by the Securities Exchange Act of 1934. Section 16(a) of the Act requires that open market

³ We do not incorporate inside trading into the VAR on purpose, because our null hypothesis is that inside trading is not informative.

trades by corporate insiders be reported to SEC within 10 days after the end of month in which they took place. For the purposes of this reporting requirement, “corporate insiders” include officers with decision making authorities over the operations of the company (CEOs, CFOs, other officers, presidents, vice-presidents etc), all members of the board of directors, and beneficial owners of more than 10% of the company’s stock. These reports filed on the SEC’s Form 3, 4 and 5 are the source of insider trading data. From the reported transactions we exclude all transactions that are less than 100 shares and only focus on open market purchases and sales by insiders.

Using the ORS data we classify insiders into three groups. The first group, Management, includes Chairmen of the board, CEO, CFO, Officers, Directors, Presidents, and Vice-Presidents and is assumed to have direct access to information about the firm’s future prospects. ‘Large shareholders’ are that who are not management but owns 10% or more of shares and are assumed to have no direct access to inside information. The third group ‘others’ are all investors who are required to report their trades to SEC but are neither managers nor large shareholders.

We define a measure of aggregate insider trading activity, IT in the following manner. For each quarter in our sample from January 1975 to December 2000 we designate a firm to be an insider buy (sell) if the number of insiders buying (selling) is greater than the number of insiders selling (buying) in that month. For each quarter IT is defined as the fraction of buys to total number of buys and sells in that quarter.

In Table 1 we present summary statistics of the trading behavior of insiders during our sampled period. On average, for the total sample, there are 1.017 insider buying stocks in their firm per quarter and 1.118 insiders selling. For the management group,

there are 0.778 number of buy per quarter and 1.432 number of sells. For the large shareholders group, there are 1.21 buys per quarter and 0.794 sells. When we look at the trading behavior across the size of the firms we notice a monotonic decrease in buys and a monotonic increase in sales for the management group. Buys decrease from .977 per quarter in the small firms to .622 per quarter in the large firms for the management group. Sales range from .796 per quarter in the small firms to 2.122 per quarter in the large firms. These results are in line with previous evidence of insiders buying more heavily in smaller firms and selling heavily in larger firms (Seyhun (1986), Rozeff and Zaman (1988)).

5. Results

The evidence presented in this section uses a VAR model to examine the relationship between aggregate insider trading and aggregate market return. We regress realized market excess returns (defined as the value-weighted return minus three month T-Bill rates) and its three estimated components (one-period expected market excess return, cash flow news and negative of discount rate news) individually on lagged values of aggregate insider trading measure, IT. The return decomposition is based on the VAR system in equation (7). We report the estimates, t-statistics, adjusted R^2 , sum of γ (β) in Panel A (B), and Granger causality test. T-statistics are computed using Newey-West heteroskedastic-robust standard errors with 5 lags, and are list below each estimate in parentheses. F-test is the Granger Causality test that all lagged inside trading (return) are zero in Panel A (B). The P-value is listed below the F-test in bracket. Results are reported in Table 2. In Panel A we report results for all insiders. First row of Panel A

shows that trading by all insiders has no explanatory power in explaining the variation in realized market returns. The F-statistic is 5.386 with a p-value of 0.146. Note that the F-statistic is used for the Granger-causality test of whether the sum of the lagged coefficients explains the variation in the independent variable. Furthermore, none of the individual lagged coefficients are significant. The second, third and fourth rows report results when the realized excess return is decomposed into one-period expected return, cash flow news and the discount rate news. The F-statistic for the expected return, cash flow news and discount rate news are 6.145, 12.686, and 2.362 with p-values of 0.105, 0.005 and 0.501 respectively. Our results suggest the following. When insiders are viewed as a monolithic group, their trading has no effect on realized market return. However, if we decompose the realized return into three components, insider trading is positively related to future aggregate cash flow news. In other words, insider trading can explain variations in realized market return which is due to future unexpected cash flow news effect. Also, the one-quarter lagged IT has a coefficient of 0.052 with a t-statistic of 1.56 for the cash flow news regression and the one-quarter-lagged IT coefficient for the NEWS regression (sum of cash flow news and discount rate news) has a co-efficient of 0.062 with a t-statistic of 2.29. These results suggest that sum of unexpected cash flow news and discount rate news experience significant positive shocks subsequent to insiders buying stocks in their firm. For the other lagged IT the co-efficients are not significant. Panel B reports results for the Management group. Here the effect of insider trading on is more pronounced. The F-statistics for realized return, expected return and cash flow news are 7.926, 10.335, and 17.651 which are all significant at the 5 per cent level. The one quarter lagged coefficients of IT for the realized return, expected return, cash flow news,

discount rate news and NEWS regressions are 0.029, 0.008, 0.039, -0.002 and 0.04 with t-statistics of 1.64, 2.32, 2.82, -0.26 and 2.85 respectively. This provides strong evidence of insiders who are directly related to the day-to-day activities of the firms is better able to predict aggregate market return. The 2-quarter and 3-quarter lagged IT coefficients are not significant. Furthermore, our results show that trading by this group of insiders is more likely to be related to unexpected future cash flow news. If insiders are buying we document a significant positive relation with cash flow news one quarter later. These results are quite different than what Chowdhury, Howe and Lin (1993) and Lakonishok and Lee (2001) claim in their paper.

Panel C reports results for the large shareholders group. F-statistics for expected return and cash flow news are marginally significant and the lagged coefficient for IT is only significant in the NEWS regression. For this group there is marginal evidence that trading is positively related to future unexpected news. Also, the 2 quarter and 3 quarter lagged coefficients for IT are all insignificant.

These results are quite different than what Chowdhury, Howe and Lin (1993) and Lakonishok and Lee(2001) claim in their papers. Both papers conclude that insiders' trade are more likely to be a function of contrarian strategy. Our results, however, show that insider trading is more likely to be based on managers' ability to time the market based on superior information. Table 2 provides evidence that insider trading is related to future cash flow news and hence is more likely to be based on the managers ability to predict market wide activities.

In Table 3 we report results of regressions between insider trading and realized market return and the components of realized market return. The motivation here is to analyze

whether it is the market wide activity that is driving insider trading and thereby supporting Chowdhury, Howe and Lin (1993) assertion of contrarian strategy. In this part of our analysis, we are more interested in the relation between insider trading and the lagged values of one-period expected market excess returns.

In panel A, like before we report results for the overall group of insiders. The F-statistics are 4.35, 3.63, 3.56, 4.16 and 5.96 when realized market excess returns, expected market excess returns, cash flow news, discount rate news and total news are regressed on lagged values of insider trading, respectively. None of the coefficients of the insider trading variable IT is significant. The co-efficient of one-quarter lagged IT is 2.034 with a t-statistic of 0.73 in the expected return regression. Panel A suggests a lack of evidence in support of the contrarian strategy. In Panel B and Panel C we report results for the Management group and the large shareholders group. Here again the evidence does not support the contrarian strategy. The co-efficient for lagged IT in the expected return regressions are 1.275 with a t-statistic of 0.25 and 1.292 with a t-statistic of 0.62 for the Management and Large shareholders group respectively. Also, none of the F-statistics are significant. Evidence presented in this table clearly shows that aggregate market returns does not cause insider trading and insider trading is not a manifestation of the contrarian strategy.

6. Firm Size, Information Uncertainty and Aggregate Insider Trading

Jiang, Lee and Zhang (2004) define information uncertainty as the degree to which a firm's value can be estimated by the most knowledgeable investors at reasonable costs. Using this definition, high information uncertainty firms would be those firms whose

expected cash flows may be difficult to estimate due to their environment or nature of operations etc. These firms are likely to have high information acquisition costs and their fundamental values are more likely to be unreliable and volatile. If aggregate insider trading is driven by the contrarian strategy then insiders are more likely to trade in high information uncertainty firms as these are more likely to have current market values deviating from the 'true' fundamental values. On the other hand, low information uncertainty firms are more likely to have market values equal to the fundamental values. Insider trading in these types of firms are more likely to be a manifestation of the insiders ability to predict aggregate market return based on managerial timing rather than contrarian strategy.

To the extent that small firms have high information acquisition costs and are likely to be followed by fewer analysts we use firm size as a proxy for information uncertainty. We repeat our earlier analyses on smaller firms and larger firms but confine it to the Management group of insiders. In table 4 we report regression results for the two groups- small firms and large firms. Realized market excess returns and its components are regressed on lagged values of IT for the small firm and large firm samples. Panel A reports results for small firms. The F-statistics for realized return, expected return, cash flow news and discount rate news regressions are 7.24, 6.294, 10.156 and 5.098 respectively. In Panel B results for large firms are reported. Here the F-statistics for realized returns, expected return, cash flow news and discount rate news are 7.324, 4.77, 15.365 and 2.311 respectively. The results suggest that for both small and large firms, insider trading is positively related and predicts the realized market returns, even though the results are marginally significant. There is a negative, marginally significant relation

between insider trading and expected return for the small firms. For both type of firms insider trading and cash flow news are positively and significantly related suggesting that insiders are able to predict future cash flow news in both cases. However, for large firms the co-efficient of one-quarter lagged value of IT is significant (co-efficient 0.055, t-statistic 2.74) while for smaller firms the third quarter coefficient is significant (co-efficient 0.096, t-statistic 2.39). For both groups insider trading is not related to discount rate news.

Table 5 reports results when IT is regressed on lagged values of realized returns, expected returns, cash flow news, discount rate news and total news for both the small firm group and the large firm group. In Panel A, for small firms the F-statistics for the regressions of realized returns, expected returns, cash flow news, discount rate news and total news are 4.07, 15.68, 3.7, 2.35, and 6.56 respectively. The only regression suggesting causality is when IT is regressed on expected returns. The evidence suggests that expected return is negatively related to insider trading. In Panel B none of the regressions are significant for the large firms.

7. Conclusion

Evidence from recent research has separately shown that insiders are able to time the market on the basis of contrarian beliefs (e.g., Rozeff and Zaman, 1998), on the basis of superior knowledge about future cash flow news (e.g., Ke et al., 2003). Piotroski and Roulstone (2005) document that insiders trade on the basis of both contrarian beliefs and superior knowledge. In this study we examine the ability of aggregate insider trading to predict market-wide movement using return decomposition in a vector autoregressive

(VAR) model framework. Following earlier research we decompose market returns into expected return, unexpected cash flow news and unexpected discount rate news. Such decomposition enables us to identify the source of predictability of aggregate insider trading. We argue that if insiders are trading on the basis of superior information then aggregate insider trading are more likely to be positively related to unexpected cash flow news. On the other hand, if these trades are a result of contrarian beliefs then insider trading should be negatively related to expected return. We find strong evidence that aggregate insider trading is positively related to unexpected cash flow news for all types of insiders. When we partition our sample based on insiders who are more likely to have access to performance related information these results are much stronger and significant. We also examine whether aggregate insider trading is in response to market expectations. We find no evidence of aggregate insider trading is caused by market expectations. Our results strongly suggest that insiders are able to predict market return because of having superior information about future cash flow news. In other words the market timing ability of aggregate insider trading is due to managerial timing and not due to contrarian strategy.

To further reinforce our results we classify firms into high information uncertainty and low information uncertainty firm and use firm size as a proxy for information uncertainty. If aggregate insider trading is due to contrarian strategy then insiders are more likely to trade in small firms and if insiders trade due to managerial timing then these trades should be concentrated on larger firms. Our results reveal that regardless of size the predictive ability of aggregate insider trading is due to managerial timing.

The fact that insider trading is due to managerial timing has an important implication. Given that insider trades are driven by superior information aggregate insider trading should therefore be construed as a leading indicator of market wide activities. Furthermore, such trading by insiders will drive prices towards fundamental values.

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Table 1
Summary Statistics

This table summarizes the statistics of insider trading for all open market purchases and sales of NYSE/AMEX and Nasdaq CRSP- and Compustat-listed common shares (CRSP share code 10 or 11) during 1978:Q1 to 2000:Q4. We exclude all option transactions and transactions less than 100 shares. We define “Management” as CEOs, CFOs, chairmen of the board, directors, officers, presidents, and vice presidents. “Large shareholders” are those who own more than 10% of shares and are not in management. “Others” are all those who are required to report their trading to the SEC but neither managers nor large shareholders. We report average quarterly number of trades per firm of our sample. Large, medium, and small firms are firms based on the sample firms’ quintile cutoff points at the market value in previous quarter.

	Management		Large shareholders		Others		Total	
	Buy	Sales	Buy	Sales	Buy	Sales	Buy	Sales
All	0.778	1.432	1.210	0.794	1.121	1.108	1.017	1.118
Small firms	0.977	0.796	1.203	0.562	1.182	0.638	1.110	0.670
Medium firms	0.748	1.332	1.155	0.780	1.064	1.080	0.973	1.062
Large firms	0.622	2.122	1.276	1.053	1.113	1.629	0.967	1.621

Table 2
Relation Between Insider Trading and Components of Market Returns as Evidence of Managerial Timing

This table shows the results of the regressions between insider trading and market return (its components). IT denotes the inside trading in equation 3, R_t denotes the realized market excess return, $E_{t-1}[R_t]$ denotes the expected market excess return, CFN_t denotes the cash flow news, DRN_t denotes the discount rate news, $NEWS_t$ denotes the sum of cash flow news and discount rate news and D is a dummy variable to control for seasonality. The return decomposition is based on the VAR system in equation 7. We report the estimates, t-statistics, adjusted R^2 , sum of γ (β) in Panel A (B), and Granger causality test. T-statistics are computed using Newey-West heteroskedastic-robust standard errors with 5 lags, and are list below each estimate in parentheses. F-test is the Granger Causality test that all lagged inside trading (return) are zero in Panel A (B). The P-value is listed below the F-test in bracket.

$$X_t = a + \sum_{i=1}^3 \beta_i X_{t-i} + \sum_{i=1}^3 \gamma_i IT_{t-i} + \sum_{k=1}^4 \phi_k D_k + e_t$$

Panel A: All Insiders						
	IT_{t-1}	IT_{t-2}	IT_{t-3}	R^2	$\sum_{i=1}^3 \gamma_i$	F-test
R_t	0.042 (1.195)	0.004 (0.091)	0.036 (0.929)	0.003	0.081	5.386 [0.146]
$E_{t-1}[R_t]$	0.015 (1.796)	-0.013 (-1.733)	-0.006 (-1.401)	0.843	-0.004	6.145 [0.105]
CFN_t	0.052 (1.561)	0.031 (0.727)	0.044 (1.159)	0.065	0.127	12.686 [0.005]
$-DRN_t$	0.007 (0.393)	0.001 (0.076)	0.009 (0.601)	0.042	0.018	2.362 [0.501]
$NEWS_t$	0.062 (2.294)	0.018 (0.577)	0.059 (1.555)	0.104	0.139	17.770 [0.000]
Panel B: Management						
R_t	0.029 (1.637)	0.007 (0.394)	0.014 (0.861)	0.010	0.049	7.926 [0.048]
$E_{t-1}[R_t]$	0.008 (2.323)	-0.008 (-2.892)	-0.002 (-0.973)	0.845	-0.002	10.335 [0.016]
CFN_t	0.039 (2.823)	0.023 (1.283)	0.012 (0.748)	0.077	0.074	17.651 [0.001]
DRN_t	-0.002 (-0.260)	-0.001 (-0.156)	0.012 (1.291)	0.059	0.009	2.584 [0.460]
$NEWS_t$	0.040 (2.850)	0.014 (0.866)	0.028 (1.753)	0.109	0.082	25.502 [0.000]
Panel C: Large Shareholders						
R_t	0.035 (0.910)	-0.023 (-0.434)	0.056 (1.148)	-0.014	0.068	3.487 [0.322]
$E_{t-1}[R_t]$	0.016 (1.824)	-0.010 (-1.095)	-0.009 (-1.692)	0.840	-0.003	7.459 [0.059]
CFN_t	0.040 (0.969)	0.016 (0.271)	0.065 (1.419)	0.039	0.120	8.251 [0.041]
DRN_t	0.012 (0.578)	-0.001 (-0.031)	0.008 (0.485)	0.044	0.020	2.404 [0.493]
$NEWS_t$	0.058 (1.953)	-0.004 (-0.083)	0.079 (1.658)	0.079	0.133	13.028 [0.005]

Table 3 Relation Between Insider Trading and Components of Market Returns as Evidence of Contrarian Strategy

This table shows the results of the regressions between inside trading and market return (its components). IT denotes the inside trading in equation 5, R_t denotes the realized market excess return, $E_{t-1}[R_t]$ denotes the expected market excess return, CFN denotes the cash flow news, DRN denotes the discount rate news, $NEWS$ denotes the sum of cash flow news and discount rate news and D is a dummy variable to control for seasonality. The return decomposition is based on the VAR system in equation 7. We report the estimates, t-statistics, adjusted R^2 , sum of γ (β) in Panel A (B), and Granger causality test. T-statistics are computed using Newey-West heteroskedastic-robust standard errors with 5 lags, and are list below each estimate in parentheses. F-test is the Granger Causality test that all lagged inside trading (return) are zero in Panel A (B). The P-value is listed below the F-test in bracket.

$$IT_t = a + \sum_{i=1}^3 \beta_i X_{t-i} + \sum_{i=1}^3 \gamma_i IT_{t-i} + \sum_{k=1}^4 \phi_k D_k + e_t$$

Panel A: All Insiders						
	X_{t-1}	X_{t-2}	X_{t-3}	R^2	$\sum_{i=1}^4 \gamma_i$	F-test
$X=R_t$	-0.003 (-0.007)	-0.244 (-0.412)	0.520 (1.806)	0.355	0.273	4.353 [0.226]
$X=E_{t-1}[R_t]$	2.034 (0.730)	-3.343 (-1.174)	-0.560 (-0.299)	0.364	-1.869	3.637 [0.303]
$X=CFN_t$	0.273 (0.850)	-0.284 (-0.502)	0.436 (1.803)	0.356	0.425	3.564 [0.313]
$X=-DRN_t$	-0.820 (-0.861)	1.067 (1.192)	1.697 (1.992)	0.379	1.945	4.156 [0.245]
$X=NEWS_t$	0.059 (0.170)	-0.153 (-0.305)	0.610 (2.223)	0.363	0.516	5.956 [0.114]
Panel B: Managemnet						
$X=R_t$	0.700 (0.851)	-0.370 (-0.369)	1.158 (1.869)	0.280	1.488	3.565 [0.312]
$X=E_{t-1}[R_t]$	1.275 (0.248)	-5.214 (-1.123)	0.208 (0.063)	0.276	-3.731	4.752 [0.191]
$X=CFN_t$	1.120 (1.418)	-0.494 (-0.505)	0.858 (1.698)	0.290	1.483	4.587 [0.205]
$X=-DRN_t$	-0.813 (-0.526)	2.159 (1.177)	3.238 (1.890)	0.288	4.585	3.691 [0.297]
$X=NEWS_t$	0.809 (1.045)	-0.286 (-0.319)	1.196 (2.100)	0.294	1.720	4.942 [0.176]
Panel C: Large Shareholders						
$X=R_t$	-0.067 (-0.195)	0.003 (0.005)	0.463 (1.765)	0.401	0.399	4.036 [0.258]
$X=E_{t-1}[R_t]$	1.292 (0.624)	-1.788 (-0.767)	-1.216 (-0.701)	0.407	-1.712	2.183 [0.535]
$X=CFN_t$	0.091 (0.327)	-0.045 (-0.100)	0.355 (1.522)	0.395	0.401	3.007 [0.391]
$X=-DRN_t$	-0.335 (-0.388)	1.071 (1.473)	1.653 (2.100)	0.425	2.389	4.807 [0.186]
$X=NEWS_t$	-0.046 (-0.161)	0.051 (0.116)	0.522 (1.875)	0.410	0.528	5.448 [0.142]

Table 4
Relation Between Insider Trading of Different Size Portfolios and Components
of Market Returns as Evidence of Managerial Timing

This table shows the results of the regressions between inside trading and market return (its components). IT denotes the inside trading in equation 3, R_t denotes the realized market excess return, $E_{t-1}[R_t]$ denotes the expected market excess return, CFN denotes the cash flow news, DRN denotes the discount rate news, $NEWS$ denotes the sum of cash flow news and discount rate news and D is a dummy variable to control for seasonality. The return decomposition is based on the VAR system in equation 7. We report the estimates, t-statistics, adjusted R^2 , sum of γ (β) in Panel A (B), and Granger causality test. T-statistics are computed using Newey-West heteroskedastic-robust standard errors with 5 lags, and are list below each estimate in parentheses. F-test is the Granger Causality test that all lagged inside trading (return) are zero in Panel A (B). The P-value is listed below the F-test in bracket.

$$X_t = a + \sum_{i=1}^3 \beta_i X_{t-i} + \sum_{i=1}^3 \gamma_i IT_{t-i} + \sum_{k=1}^4 \phi_k D_k + e_t$$

Panel A: small size

	IT_{t-1}	IT_{t-2}	IT_{t-3}	R^2	$\sum_{i=1}^4 \gamma_i$	F-test
R_t	0.031 (0.788)	-0.010 (-0.230)	0.072 (2.079)	0.015	0.093	7.240 [0.065]
$E_{t-1}[R_t]$	0.010 (1.131)	-0.010 (-1.217)	-0.006 (-1.204)	0.834	-0.006	6.294 [0.098]
CFN_t	0.047 (1.441)	0.012 (0.256)	0.096 (2.389)	0.087	0.155	10.156 [0.017]
$-DRN_t$	0.022 (1.514)	0.006 (0.297)	-0.006 (-0.418)	0.066	0.022	5.098 [0.165]
$NEWS_t$	0.071 (2.314)	0.005 (0.153)	0.094 (2.623)	0.130	0.170	16.049 [0.001]

Panel B: Big size

R_t	0.044 (2.350)	0.005 (0.239)	0.023 (0.926)	0.018	0.072	7.324 [0.062]
$E_{t-1}[R_t]$	0.009 (1.627)	-0.009 (-1.708)	-0.001 (-0.450)	0.835	-0.001	4.770 [0.189]
CFN_t	0.055 (2.740)	0.017 (0.755)	0.017 (0.674)	0.044	0.089	15.365 [0.002]
DRN_t	0.000 (-0.008)	0.003 (0.246)	0.012 (0.794)	0.049	0.015	2.311 [0.510]
$NEWS_t$	0.055 (3.723)	0.011 (0.647)	0.032 (1.273)	0.075	0.099	18.942 [0.000]

Table 5
Relation Between Insider Trading of Different Size Portfolios and Components of Market Returns as Evidence of Contrarian strategy

This table shows the results of the regressions between inside trading and market return (its components). IT denotes the inside trading in equation 5, R_t denotes the realized market excess return, $E_{t-1}[R_t]$ denotes the expected market excess return, CFN denotes the cash flow news, DRN denotes the discount rate news, $NEWS$ denotes the sum of cash flow news and discount rate news and D is a dummy variable to control for seasonality. The return decomposition is based on the VAR system in equation 7. We report the estimates, t-statistics, adjusted R^2 , sum of γ (β) in Panel A (B), and Granger causality test. T-statistics are computed using Newey-West heteroskedastic-robust standard errors with 5 lags, and are list below each estimate in parentheses. F-test is the Granger Causality test that all lagged inside trading (return) are zero in Panel A (B). The P-value is listed below the F-test in bracket.

$$IT_t = a + \sum_{i=1}^3 \beta_i X_{t-i} + \sum_{i=1}^3 \gamma_i IT_{t-i} + \sum_{k=1}^4 \phi_k D_k + e_t$$

Panel A: small size						
	IT_{t-1}	IT_{t-2}	IT_{t-3}	R^2	$\sum_{i=1}^4 \gamma_i$	F-test
$X=R_t$	-0.453 (-1.252)	-0.468 (-0.842)	0.137 (0.556)	0.353	-0.784	4.070 [0.254]
$X=E_{t-1}[R_t]$	1.831 (0.696)	-2.330 (-0.977)	-3.134 (-1.701)	0.419	-3.632	15.678 [0.001]
$X=CFN_t$	-0.041 (-0.171)	-0.237 (-0.462)	0.303 (1.911)	0.331	0.025	3.703 [0.295]
$X=-DRN_t$	-0.771 (-0.929)	0.787 (0.969)	1.498 (1.440)	0.359	1.514	2.350 [0.503]
$X=NEWS_t$	-0.196 (-0.663)	-0.163 (-0.337)	0.464 (2.359)	0.343	0.106	6.556 [0.087]
Panel B: Big size						
$X=R_t$	0.163 (0.337)	-0.435 (-0.554)	0.640 (1.841)	0.270	0.367	3.963 [0.265]
$X=E_{t-1}[R_t]$	3.726 (0.885)	-4.857 (-1.100)	0.294 (0.133)	0.264	-0.837	1.883 [0.597]
$X=CFN_t$	0.495 (1.410)	-0.730 (-0.935)	0.534 (1.632)	0.285	0.300	4.054 [0.256]
$X=-DRN_t$	-1.411 (-1.079)	1.963 (1.667)	1.901 (1.731)	0.308	2.452	3.828 [0.281]
$X=NEWS_t$	0.133 (0.315)	-0.417 (-0.601)	0.685 (2.026)	0.271	0.401	4.334 [0.228]